Course: **GRC103 — Risk Assessment and Management Techniques**
Assessment: **Advanced Cybersecurity Risk Quantification and Control Investment Analysis**
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Phase 1: Risk Exposure Calculation

Task 1: Calculate Annualised Loss Expectancy(ALE)

Critical Finding 1: API Authentication Bypass

SLE= $5 000 000
ARO = 15% > 0.15
ALE = **SLE** * **ARO**
= $5 000 000 * 0.15
= $750 000

Critical Finding 2: Database Injection Vulnerability

SLE = 500 000 * $250 = 125 000 000
ARO = 25%
ALE = **SLE** * **ARO**
= 125 000 000 * 0.25
= 31 250 000

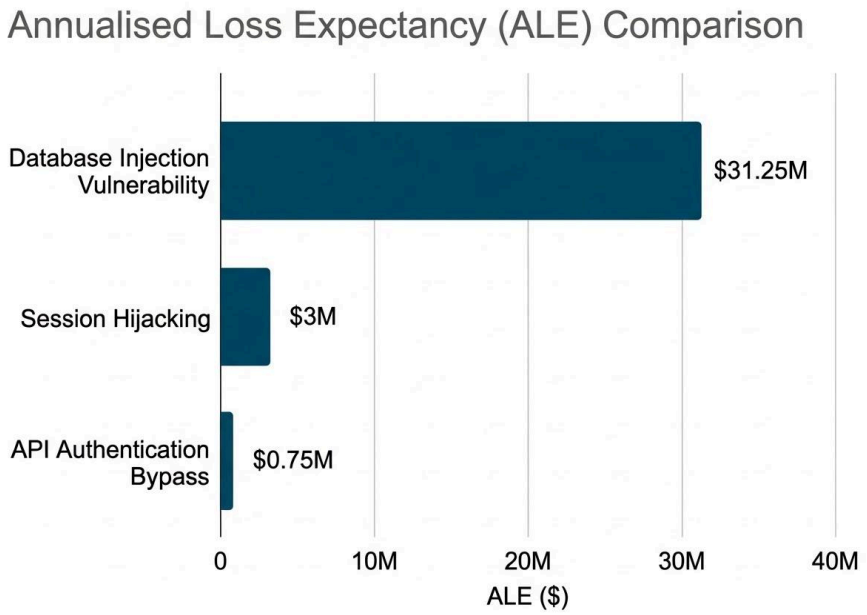
Critical Finding 3: Session Hijacking

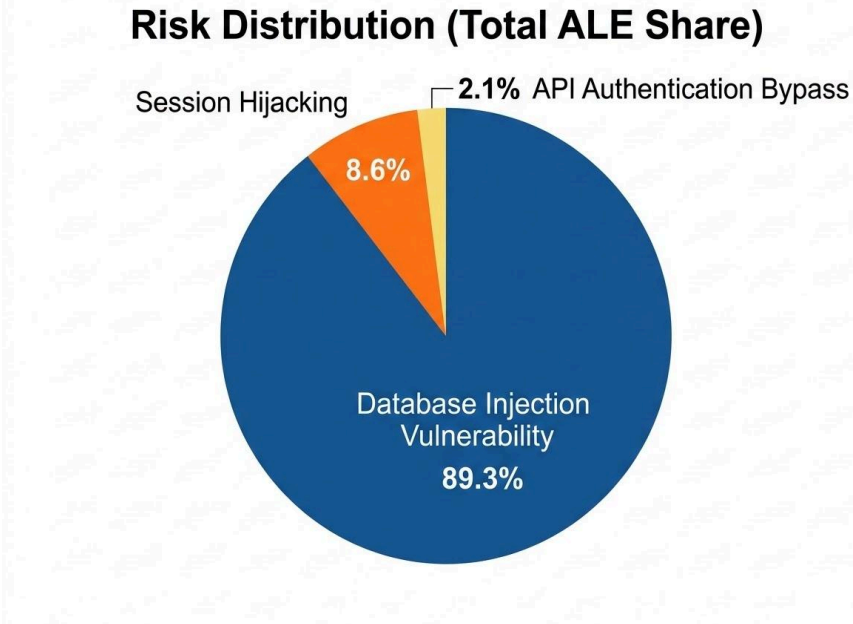
SLE = 5000 * 1500
ARO = 40%
ALE = **SLE** * **ARO**
= 7 500 000 * 0.40
= 3 000 000

The Risk Matrix

Priority	Vulnerability	ARO	SLE($)	ALE ($)	Level
1	Database Injection Vulnerability	0.25(25%)	125 000 000	31, 250, 000	
2	Session Hijacking	0.40(40%)	7 500 000	3, 000, 000	
3	API Authentication Bypass	0.52 (15%)	5 000 000	750, 000	

Task 3: Create Risk Visualization





Risk Heat Map

	Low Impact	Medium Impact	High Impact
High Probability	**Medium**	**High** **SESSION HIJACKING**	**High**
Medium Probability	**Low**	**Medium**	**High** **DATABASE INJECTION**
Low Probability	**Low**	**Medium** **API BYPASS**	**High**

Phase 2: Control Evaluation

1. ### Advanced API Security Gateway

Initial Investment - $350, 000
Annual Maintenance - $50, 000
Risk Reduction -
Total year 1 Cost: $350, 000 + $50, 000 = $400, 000
$750, 000 * 0.90 = $675, 000

Return on Investment (ROI) -
ROI = $675, 000 - $400, 000 / $400, 000 * 100 = $275, 000 / $400, 000 * 100 = 68,75%

Payback Period -
Net Annual Benefit = $675, 000 - $50, 000 = $625, 000
Payback Period = $350, 000/ $625, 000 = 0.56 years (=6.7 months)

2. ### Web Application Firewall

Initial Investment - $150, 000
Annual Maintenance - $25, 000
Total year 1 Cost: $150, 000 + $25, 000 = $175, 000

Risk Reduction
$31, 250, 00 * 0,75 = $23, 437, 500
Return on Investment (ROI) -
ROI = $23, 437, 500 - $175, 000 / $175, 000 * 100 = $23, 262, 500 / $175, 000 * 100 =13,292.86%

Payback Period -
Net Annual Benefit = $23, 437, 500 - $25, 000 = $23, 412, 500
Payback Period = $150, 000/ $23, 412, 500 = 0.0064 years (= 2.3 days)

3. Multi-Factor Authentication Enhancement

Initial Investment - $200, 000
Annual Maintenance - $30, 000
Total year 1 Cost: $200, 000 + $30, 000 = $230, 000

Risk Reduction(95%)
$31, 250, 00 * 0,75 = $23, 437, 500

Return on Investment (ROI) -
ROI = $2,850,000 - $230, 000 / $230, 000 * 100 = $2, 620, 000 / $230, 000 * 100 =1,139.13%

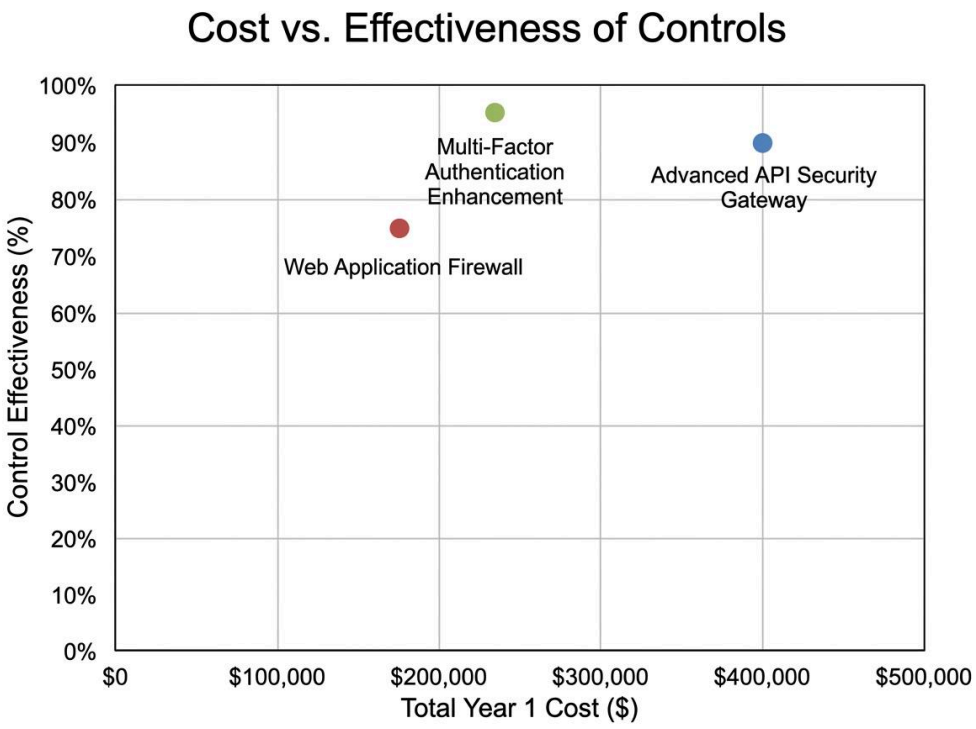
Payback Period -
Net Annual Benefit = $2,850, 000 - $30, 000 = $2, 820, 000
Payback Period = $200, 000/ $2, 820, 000 = 0.071 years (= 26 days)

Summary Table

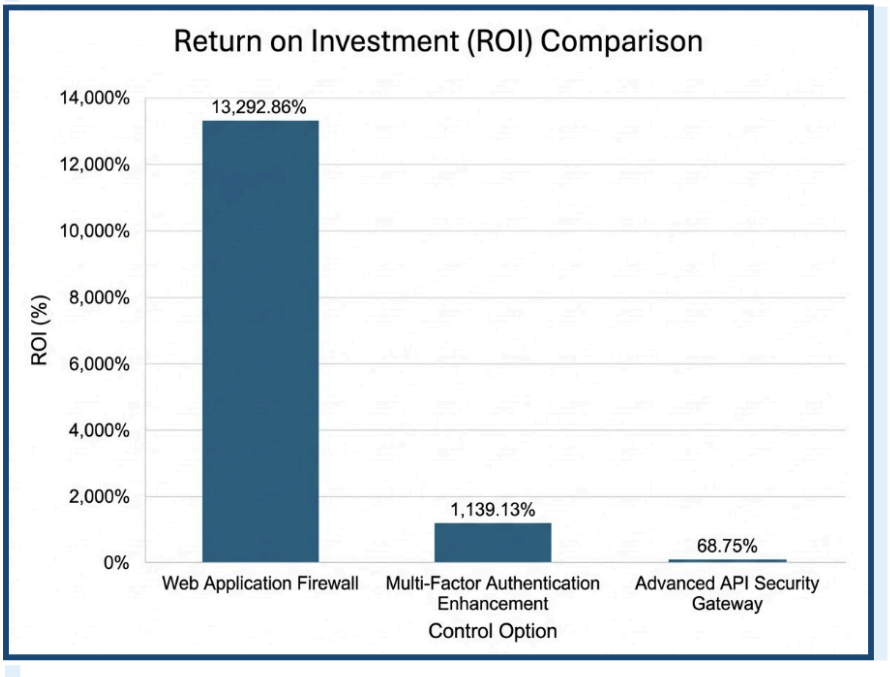
Control Option	Target Vulnerability	Initial Investment ($)	Annual Maintenance	Total Year 1 Cost	Risk Reduction	ROI(%)	Payback Period
Advanced Api Security Gateway	API Authenticatio n Bypass	350,000	50, 000	400, 000	675, 000	68,75	0.56 year (6.7 mos)
Web Application Firewall	Database Injection	150, 000	25, 000	175, 000	23, 437, 500	13, 292.86	0.00064 years (2.3 days)
Muti-Factor Authenticati on (MFA)	Session Hijacking	200, 000	30, 000	230, 000	2, 850, 000	1,139.13	0.071 years (25.9 days)

Task 5: Control Selection Analysis

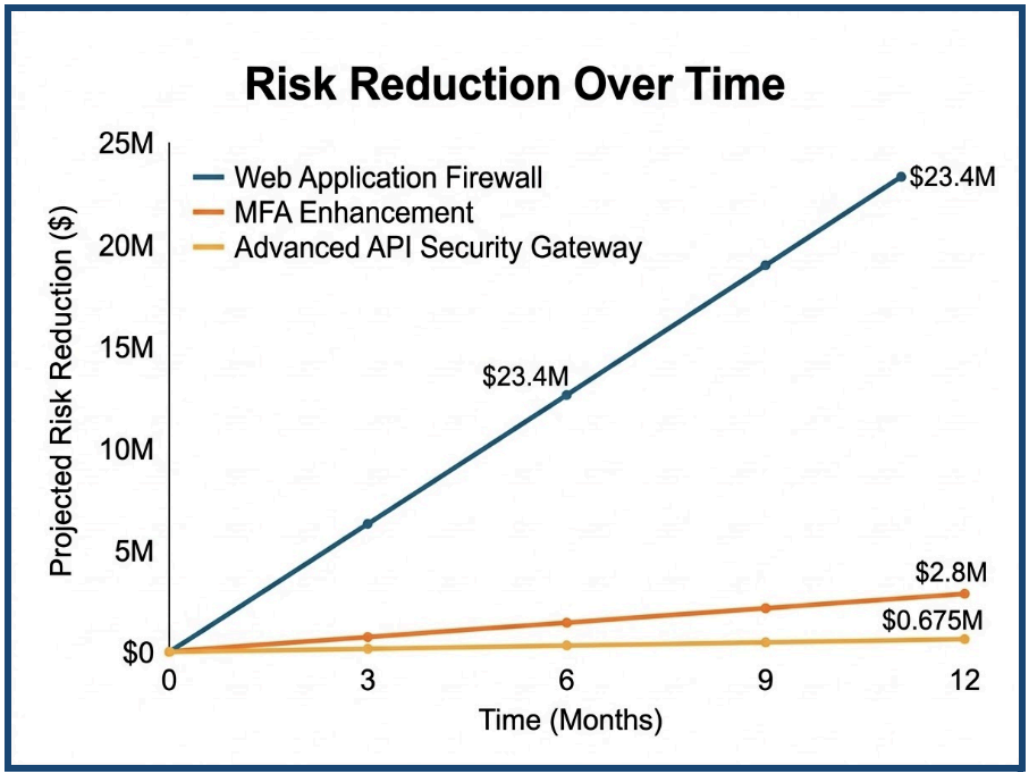
Scatter Plot Graph



Bar Chart



Line Graph



Phase 3: Executive Reporting

Executive Briefing: Cybersecurity Risk Quantification

Target System: SecureMobile Banking Application (500,000 Active Users)
Prepared For: Chief Information Security Officer (CISO) & Chief Financial Officer (CFO)
Prepared By: Lead GRC Analyst

1. Executive Summary

A comprehensive quantitative risk assessment of the *SecureMobile* platform identified **$35,000,000** in total Annualized Loss Expectancy (ALE) across three critical vulnerability vectors. By investing **$700,000** in targeted security controls (with **$105,000/year** in ongoing maintenance), SecureBank Financial can eliminate **$26,962,500** in annual risk exposure—achieving an overall first-year net risk reduction of **$26,157,500**.

2. Top 3 Risks & Financial Impact

Rank & Vulnerability	Exploitation Probability (ARO)	Single Loss Expectancy (SLE)($)	Annualized Loss Expectancy (ALE)($)	Risk Priority Level
Database Injection	25%	125, 000, 000	32, 250, 000	Critical Priority 1
Session Hijacking	40%	7, 500, 000	3, 000 000	High Priority 2
API Authentication Bypass	15%	5, 000 000	750, 000	Meidum Priority 3
Total Baseline Exposure	-	-	35, 000, 000	-

Key Takeway: Database Injection accounts for **89.3%** of the total organizational financial exposure due to the scale of customer profiles at risk (500,000 profiles at $\$250$/record).

3. Recommended Controls, Financial Justification & ROI

Recommended Control Option	Target Vulnerability	Initial Investment ($)	Annual Maintenance	Total Year 1 Cost	Expected Risk Reduction	ROI(%)	Payback Period
Advanced Api Security Gateway	API Authenticati on Bypass	350,000	50, 000	400, 000	675, 000	68,75	0.56 year (6.7 mos)
Web Application Firewall	Database Injection	150, 000	25, 000	175, 000	23, 437, 500	13, 292.86	0.00064 years (2.3 days)
Muti-Factor Authentication (MFA)	Session Hijacking	200, 000	30, 000	230, 000	2, 850, 000	1,139.13	0.071 years (25.9 days)
Total Security Investment	All 3 Controls	700, 000	105, 000	805, 000	26, 962, 500	3, 249.38	11 days

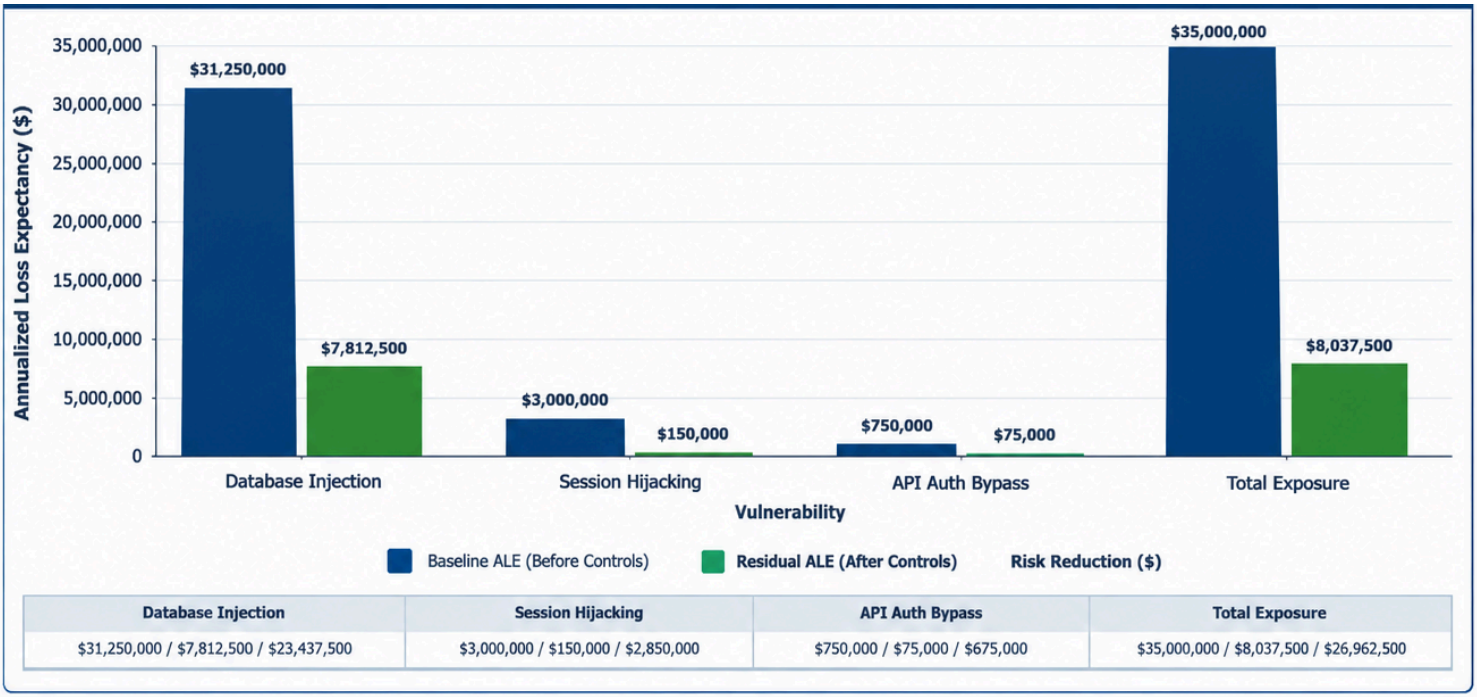
4. Post-Control Residual Risk Exposure

Baseline Financial Risk Exposure (Current ALE) = $35, 000, 000
Total Annual Risk Reduction Achieved = -26, 962, 500
Residential Risk Exposure(New ALE) = $8, 037, 500

5. **Strategic Recommendations for Leadership**

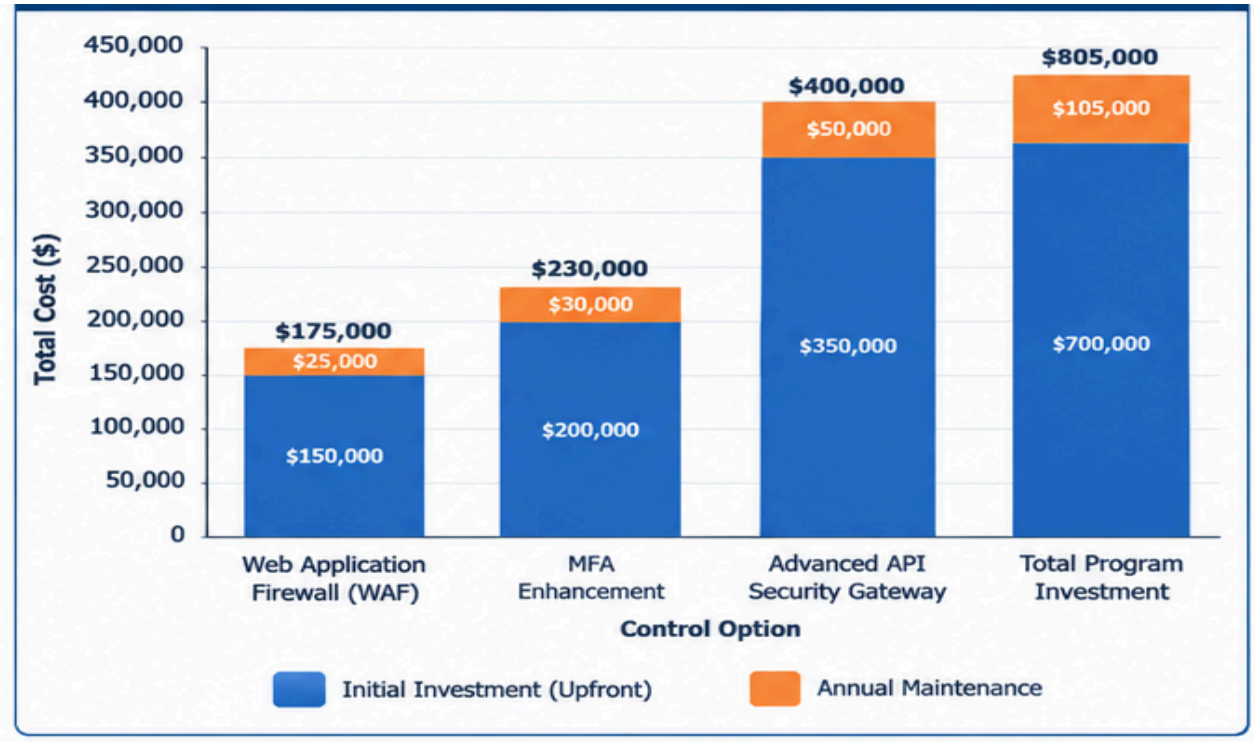
- **Immediate Execution (WAF Deployment):** Prioritize the Web Application Firewall implementation immediately. It addresses our largest vulnerability (89.3% of overall financial exposure) and pays for itself within 3 days of operational deployment.
- **Account Protection (MFA Enhancement):** Deploy multi-factor authentication to secure 5,000 simultaneous active user sessions, dropping account takeover exposure from $\$3.0\text{M}$ to $\$150,000$.
- **API Perimeter Security:** Deploy the API Security Gateway to close the 48-hour detection window and protect core transaction processing endpoints.

Risk Exposure Before/After Controls (Double Bar Chart)



Database Injection	Session Hijacking	API Auth Bypass	Total Exposure
$31,250,000 / $7,812,500 / $23,437,500	$3,000,000 / $150,000 / $2,850,000	$750,000 / $75,000 / $675,000	$35,000,000 / $8,037,500 / $26,962,500

Control Investment Breakdown (Stacked Bar Chart)



ROI Comparison Across Controls (Horizontal Bar Chart)



Risk Treatment Timeline

Summary for the chart

- **Phase 1 Priority (Risk Neutralization):** Deployment focuses heavily on WAF virtual patching during Days 1–15 to immediately neutralize Database Injection exposure (89.3% of total risk) with a 2. 3-day payback period.
- **Phase 2 Operational Scope (Identity & Application):** MFA rollout addresses Session Hijacking across all 5,000 active sessions, while permanent database code refactoring eliminates the underlying SQL injection flaw.
- **Phase 3 Governance & Assurance (Long-Term):** API Gateway integration secures transaction endpoints, backed by continuous ISO 27001/42001 ISMS alignment and an annual penetration test to verify residual risk remains below $8.04M.

Bonus Task



Summary: Combined Risk-Control Matrix

The Combined Risk-Control Matrix provides an executive overview of SecureBank Financial's cybersecurity risk exposure, proposed security controls, implementation costs, and expected financial benefits. It combines the baseline Annual Loss Expectancy (ALE) for each vulnerability with the estimated residual risk after applying the recommended controls.

The matrix identifies **Database Injection** as the largest financial exposure, with a baseline ALE of $31.25 million. The recommended Web Application Firewall (WAF) has a first-year cost of $175,000 and is estimated to reduce this exposure by $23.44 million annually. **Session Hijacking** is addressed through Multi-Factor Authentication (MFA), costing $230,000 in the first year and reducing the estimated annual loss by $2.85 million. **API Authentication Bypass** is addressed through an Advanced API Security Gateway, costing $400,000 in the first year and reducing the estimated annual loss by $675,000.

The charts below the main table visually compare the baseline and residual risk exposures, break down the initial investment and annual maintenance costs of each control, and compare their estimated Return on Investment (ROI). The budget optimisation section presents three

implementation options: deploying the WAF first, combining the WAF with MFA, or implementing all three controls for broader security coverage.

Overall, the proposed full program requires a first-year investment of $805,000 and is projected to reduce the combined ALE from $35 million to $8.0375 million, representing an estimated 77% reduction in annualized risk exposure. These figures are model-based estimates that depend on the assumed control effectiveness and risk calculations. The matrix supports management in prioritizing remediation, allocating resources according to budget constraints, and making informed decisions about cybersecurity investment.